

Spatio-Temporal Mapping and Analysis of Surface-Water Dynamics in Kopa Dam, Niger State, Nigeria, Using Sentinel-2 Level-2A and GIS Techniques

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Abstract

Surface-water extent provides a spatial measure of reservoir condition and supports water-resource monitoring where field observations are limited. This study assessed the spatio-temporal dynamics of surface water in Kopa Dam, Niger State, Nigeria, from January 2022 to January 2026 using Sentinel-2 Level-2A imagery and the Modified Normalized Difference Water Index (MNDWI). One image was selected for January of each year to maintain seasonal consistency. MNDWI was calculated using the green band (B03) and shortwave infrared band (B11), with B11 resampled to 10 m. Water and non-water reference points were used to determine image-specific thresholds of -0.01, -0.01, -0.18, -0.01, and -0.25 for 2022, 2023, 2024, 2025, and 2026, respectively. Binary water masks were generated and refined using the Majority Filter. Mapped water areas were 3.3097, 3.2929, 4.4854, 3.7591, and 4.2103 km² for 2022–2026, respectively. The largest increase occurred between 2023 and 2024 (+36.21%), while the largest decline occurred between 2024 and 2025 (-16.19%). Overall, water area increased by 0.9006 km² (27.21%) between 2022 and 2026. The linear time-series model indicated a positive tendency ($R^2 = 0.4523$). Spatial transition analysis identified 95.46 ha of water gain, 5.64 ha of water loss, 325.31 ha of persistent water, and 85.63 ha of stable non-water. The findings indicate substantial annual fluctuations rather than a uniform directional change and demonstrate the usefulness of Sentinel-2 and MNDWI for reproducible reservoir monitoring. Continued application with field water-level observations and bathymetric surveys is recommended for improved reservoir management.

Keywords: Kopa Dam, Surface Water, MNDWI, Sentinel-2, Water-Area Change, GIS.

Introduction

Reservoirs are important components of water-supply systems, irrigation networks, flood-management systems, and local economic activities. Their surface-water extent changes with inflow, rainfall, evaporation, abstraction, reservoir operation, sedimentation, and activities around the catchment. Monitoring these changes requires observations that are spatially consistent and repeatable. Satellite remote sensing provides such observations and has become an important source of information for mapping inland water extent over large and small areas. Global surface-water research has demonstrated that satellite

imagery can capture both the location and temporal changes of inland water bodies over long periods (Pekel et al., 2016).

The Sentinel-2 Multispectral Instrument provides 13 spectral bands at 10, 20, and 60 m spatial resolutions and was designed to support land and inland-water monitoring. The 10 m visible bands provide detailed spatial information, while the 20 m shortwave infrared bands provide spectral information useful for distinguishing water from surrounding land features (European Space Agency [ESA], 2026). Level-2A products provide atmospherically corrected surface reflectance and include scene-classification information, making them suitable for consistent multi-temporal surface-water analysis.

The Modified Normalized Difference Water Index was developed to improve open-water enhancement by replacing the near-infrared band in the conventional water-index formulation with a middle-infrared band. The use of shortwave infrared reflectance strengthens the spectral separation between open water and several non-water surfaces, including built-up land, vegetation, and soil (Xu, 2006). This makes MNDWI particularly relevant to reservoir environments where shoreline materials and human activities can complicate water delineation.

The suitability of a water index does not mean that one numerical threshold will work equally well for every image. Thresholds depend on the spectral characteristics of the scene, the composition of water and non-water pixels, and local environmental conditions. Ji et al. (2009) demonstrated the need to adjust water-index thresholds according to the actual image conditions. Recent Sentinel-2 research in Ethiopia also found differences in the performance of water indices across water environments, supporting careful image-specific assessment (Tesfaye & Breuer, 2024). This supports an image-specific validation approach rather than imposing a single threshold on every acquisition.

The Nigerian context further supports the need for local reservoir monitoring. Frichi et al. (2023) used satellite-derived water indices to assess changes in Kiri Dam, Adamawa State, and reported substantial changes in reservoir surface-water extent over time. Olaniyi et al. (2023) also evaluated Sentinel-2A and Landsat-based water extraction in the Ise Forest Reserve in southwestern Nigeria and found MNDWI among the effective approaches for surface-water mapping. At Suleja Dam, Samaila-Ija et al. (2014) conducted a bathymetric and volumetric assessment and reported differences between the designed and surveyed reservoir dimensions, with the authors associating the observed reservoir-bed characteristics with sedimentation and siltation processes.

Kopa Dam, also referred to as Suleja Dam in the study of Samaila-Ija et al. (2014), is located within the Suleja emirate of Niger State. Recent research identifies the dam as an important water-storage structure within the local drainage system and reports a reservoir extent of approximately 5 km². Despite its importance, a consistent five-year assessment of its surface-water extent using the same month, sensor, spatial framework, and validated MNDWI classification approach has not yet been established. Establishing such a baseline can support water-supply planning by providing a spatial record of changes in the reservoir

surface that can be considered alongside water-level and operational records when assessing water-supply reliability and reservoir-capacity conditions.

The main aim of this study is therefore to map and quantify surface-water changes in Kopa Dam between January 2022 and January 2026 using Sentinel-2 Level-2A imagery and MNDWI. The specific objectives were to delineate annual surface-water extent, determine image-specific MNDWI thresholds through reference-point validation, quantify annual and overall water-area changes, examine the temporal trend of water extent, and map spatial water gains and losses across the study period.

Remote sensing provides a practical means of monitoring surface-water extent because satellite imagery offers repeated and spatially consistent observations over large areas. Long-term Landsat analysis has demonstrated substantial temporal and spatial changes in global surface-water occurrence, confirming the value of satellite observations for monitoring water dynamics (Pekel et al., 2016). The availability of Sentinel-1, Sentinel-2, and Sentinel-3 has further expanded opportunities for inland-water monitoring, including the assessment of water extent, water level, water depth, and water quality (Zeng et al., 2023).

For reservoirs, surface-water extent provides a useful spatial indicator of changing water conditions. However, surface area should not be treated as a direct measure of storage volume because reservoir volume also depends on water depth and basin geometry. Remote sensing can therefore provide an important spatial complement to water-level and bathymetric observations rather than replacing them (Zeng et al., 2023).

Sentinel-2 is well suited to surface-water mapping because its Multispectral Instrument provides 13 spectral bands at 10, 20, and 60 m spatial resolutions. Its visible, near-infrared, and shortwave-infrared bands provide useful spectral information for separating water from surrounding land-cover types. Level-2A products provide atmospherically corrected surface reflectance, making them appropriate for multi-temporal analysis (European Space Agency [ESA], 2026).

Water indices simplify surface-water extraction by enhancing the spectral contrast between water and non-water surfaces. The Normalized Difference Water Index (NDWI) uses green and near-infrared bands, whereas the Modified Normalized Difference Water Index (MNDWI) replaces the near-infrared band with shortwave infrared. Xu (2006) developed MNDWI to improve open-water discrimination, particularly where built-up surfaces and other non-water features create confusion.

Comparative studies have shown that MNDWI can perform well for surface-water extraction, but its effectiveness is not universal. Index performance varies with environmental conditions, surrounding land cover, and the characteristics of individual water bodies (Tesfaye & Breuer, 2024). Recent Sentinel-2 studies similarly indicate that different indices can produce different results under varying landscape conditions. Therefore, selection of an appropriate water index should be accompanied by evaluation of its performance within the specific study environment.

Although MNDWI produces a continuous range of values, surface-water mapping requires a threshold to separate water from non-water pixels. The threshold selected can substantially influence the resulting water extent. Applying a universal threshold across multiple images may therefore introduce classification errors because the spectral characteristics of water and surrounding surfaces can change between acquisition dates.

Research on multitemporal Landsat-8 and Sentinel-2 imagery has demonstrated uncertainty associated with water-index thresholds and emphasized the importance of appropriate threshold selection (Reis et al., 2021). Recent studies have also shown that threshold performance varies according to environmental conditions and the characteristics of individual scenes (Tesfaye & Breuer, 2024). These findings support the use of image-specific threshold assessment when analyzing multi-temporal surface-water changes.

Multi-temporal mapping also provides information beyond individual water-area measurements. Comparing classified water masks allows annual expansion and contraction to be quantified and enables persistent water, water gain, water loss, and stable non-water areas to be identified. Such spatial transitions provide additional information about reservoir dynamics that cannot be obtained from total water area alone (Pekel et al., 2016). Studies in Nigeria have demonstrated the usefulness of satellite imagery and water indices for monitoring reservoirs and inland water bodies. Frichi et al. (2023), for example, reported substantial inter-annual surface-water changes at Kiri Dam and found MNDWI to be effective among the tested indices. Ngebe et al. (2022) also demonstrated the potential of Sentinel-2 imagery for supporting dam water-level estimation when combined with elevation and field observations. These studies show the potential of freely available satellite imagery for Nigerian water-resource monitoring.

However, existing studies differ in their sensors, observation periods, indices, thresholding procedures, and temporal frameworks. For Kopa Dam, there is limited evidence of a consistent multi-year assessment based on the same seasonal observation period and an image-specific MNDWI threshold-validation approach. This creates a need for a reproducible framework that can distinguish annual fluctuations from longer-term changes.

The present study addresses this gap by combining January Sentinel-2 Level-2A imagery from 2022 to 2026 with MNDWI, reference-point threshold validation, GIS-based water-area estimation, temporal trend analysis, and spatial transition mapping. The framework provides a consistent basis for documenting surface-water dynamics at Kopa Dam while recognizing that surface-water extent should be interpreted alongside water-level and bathymetric information when assessing reservoir storage conditions.

Materials and Methods

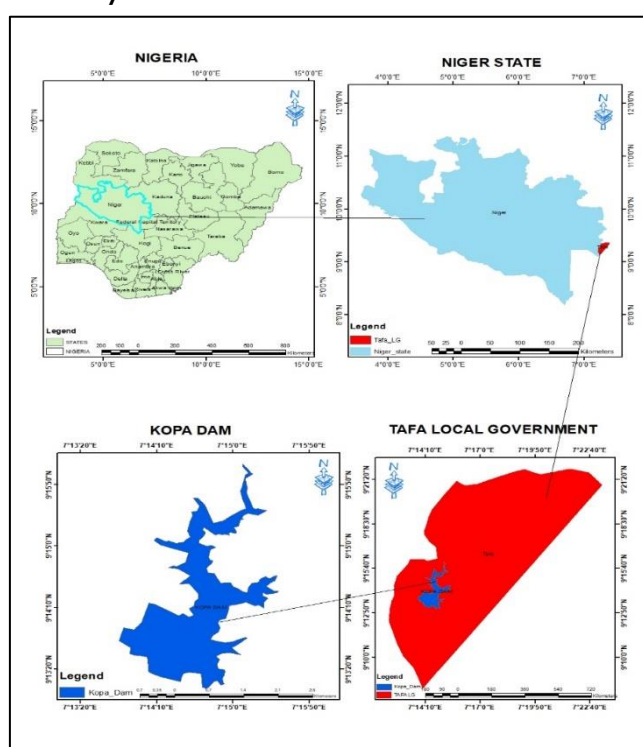
Study area

The study was conducted at Kopa Dam, situated in Tafa Local Government Area of Niger State, Nigeria, centred at approximately 9°14'10"N, 7°14'45"E. Kopa Dam is also referred to

as Suleja Dam in previous studies (Samaila-Ija et al., 2014; Umar et al., 2026). Recent research identifies the dam as an important water-storage structure within the local drainage system and reports a reservoir extent of approximately 5 km² (Umar et al., 2026). The study focuses on documenting changes in mapped surface-water extent and providing a spatial baseline that can be considered alongside water-level and operational information for reservoir monitoring.

Beyond domestic water supply, the reservoir supports irrigation, flood control, and fisheries for the surrounding communities, and its shoreline is increasingly used for recreational activities such as fishing and picnicking.

Figure 1: Map of the Study Area.



Source: Author, 2026.

Satellite Data

Five Sentinel-2 Level-2A images were used. One image was selected for January of each year from 2022 to 2026. Using the same month reduced the effect of comparing different seasonal periods. The Level-2A products contained atmospherically corrected surface-reflectance data. The analysis used B03 and B11 for MNDWI calculation. B03 has a native spatial resolution of 10 m, while B11 has a native spatial resolution of 20 m. B11 was therefore resampled to 10 m before index calculation.

The image-quality information recorded for the selected scenes is presented below.

Table 1. Sentinel-2 image quality information for the selected January scenes

Year	Cloudy pixels over land (%)	Water (%)	Vegetation (%)
2022	4.13	2.86	1.48
2023	5.43	2.83	0.40
2024	4.01	3.19	0.87
2025	3.94	3.25	1.25
2026	0.76	3.05	0.91

Source: Author, 2026.

The selected scenes were considered suitable for the analysis because the reported cloud-related proportions were analyzed and found absent within the region of interest and the images provided continuous coverage of the reservoir.

Image Preparation

The Level-2A bands were prepared in ArcGIS 10.8. The B03 and B11 rasters were extracted from the corresponding Sentinel-2 products. The B11 raster was resampled from 20 m to 10 m to match the spatial resolution of B03. The resampling operation used a 10 m output cell size and bilinear resampling algorithm. The resampled B11 raster was aligned with the B03 raster and the study-area grid. All spatial processing and maps used the WGS 1984 UTM Zone 32N coordinate reference system.

The MNDWI rasters were then calculated for each year. Raster processing was performed within the same study-area boundary to ensure that subsequent pixel counts represented the same geographic extent.

MNDWI Calculation

This study utilized MNDWI as the sole spectral index for surface-water delineation.

The index was calculated as:

$$\text{MNDWI} = (\rho_{\text{Green}} - \rho_{\text{SWIR}}) / (\rho_{\text{Green}} + \rho_{\text{SWIR}})$$

For Sentinel-2:

$$\text{MNDWI} = (B_{03} - B_{11}) / (B_{03} + B_{11})$$

Where:

B03 = green-band surface reflectance

B11 = shortwave infrared surface reflectance

The calculation was performed separately for 2022, 2023, 2024, 2025, and 2026.

MNDWI Threshold Validation

A fixed threshold was not imposed on all five images. Instead, the threshold was evaluated using reference points representing water and non-water surfaces. The reference points were manually created within the study area and labelled by visual interpretation of the corresponding Sentinel-2 scene and available ancillary information. No random, systematic, or stratified probability-sampling design was applied. A single Class field was

used, with Class 1 representing water and Class 2 representing non-water. The total number of reference points created for each year was 161 for 2022, 160 for 2023, 161 for 2024, 173 for 2025, and 151 for 2026. MNDWI values were then extracted at the reference points using the Extract Multi Values to Points tool in ArcGIS 10.8.

The reference points were assigned to two classes:

Class 1 = water

Class 2 = non-water

MNDWI values were extracted from each reference point. The minimum, maximum, mean, standard deviation, and number of observations were examined for each class.

Threshold candidates were then tested using the extracted point values. A classification field was created using the following rule:

Predicted class = 1, if $MNDWI \geq T$

Predicted class = 2, if $MNDWI < T$

where T is the tested threshold.

The final threshold was selected from the validated classification that provided the best separation between the reference water and non-water classes.

The final thresholds were:

Table 2. Image-specific MNDWI thresholds used for water delineation

Year	Selected MNDWI threshold
2022	-0.01
2023	-0.01
2024	-0.18
2025	-0.01
2026	-0.25

Source: Author, 2026.

The threshold assessment demonstrated that the appropriate threshold was scene-specific. The selected thresholds were -0.01 for 2022, -0.01 for 2023, -0.18 for 2024, -0.01 for 2025, and -0.25 for 2026. The final MNDWI thresholds showed reference-point agreement of 86.34%, 100.00%, 100.00%, 95.95%, and 100.00% for 2022, 2023, 2024, 2025, and 2026, respectively. These percentages represent agreement between the selected thresholds and the reference points used during threshold assessment. The threshold tests were used to select the final scene-specific thresholds, while the confusion matrices reported below provide the formal accuracy assessment.

A confusion matrix was prepared for each year's reference dataset by cross-tabulating the classified water/non-water values against the reference classes. The assessment used valid point records with corresponding classified values. The resulting overall accuracies ranged from 94.38% to 100.00%. The 2022 assessment recorded 95.03% overall accuracy and a Kappa coefficient of 0.900. The 2023 assessment recorded 94.38% overall accuracy and a Kappa coefficient of 0.888, with nine false negatives and no false positives. The 2024

assessment achieved 100.00% overall accuracy and a Kappa coefficient of 1.000 based on 159 valid points. The 2025 assessment recorded 96.53% overall accuracy and a Kappa coefficient of 0.930, with six false negatives and no false positives. The 2026 assessment achieved 100.00% overall accuracy and a Kappa coefficient of 1.000 based on 144 valid points. The class totals and confusion-matrix counts are presented in Table 3.

Table 3. Confusion matrices and accuracy statistics for annual water-mask validation

Year	Reference water, classified water (TP)	Reference water, classified non-water (FN)	Reference non-water, classified water (FP)	Reference non-water, classified non-water (TN)	Valid points	Overall accuracy (%)	Kappa
2022	72	5	3	81	161	95.03	0.900
2023	71	9	0	80	160	94.38	0.888
2024	80	0	0	79	159	100.00	1.000
2025	74	6	0	93	173	96.53	0.930
2026	80	0	0	64	144	100.00	1.000

Source: Author, 2026.

Binary Water-Mask Generation

Each MNDWI raster was converted into a binary water mask using the validated annual threshold.

The classification equation was:

WaterMask =
1, if MNDWI ≥ T
0, if MNDWI < T

The resulting raster contained:

- 0 = non-water
- 1 = water

The binary water masks contained isolated pixels and small classification artefacts. A Majority Filter was applied to reduce these isolated pixels and improve the spatial coherence of the water masks.

The ArcGIS Majority Filter was applied using the FOUR neighborhood rule and HALF replacement criterion. The filtered outputs were used for the final water-area calculations and subsequent change analysis.

Surface-Water Area Calculation and Annual Percentage Change

The Sentinel-2 analysis grid had a 10 m spatial resolution. Each pixel therefore represented:

- Pixel area = 10 m × 10 m
- Pixel area = 100 m²

Since:

1 km² = 1,000,000 m²
the area represented by one pixel was:

$$100 / 1,000,000 = 0.0001 \text{ km}^2$$

Annual water area was calculated as:

$$A = N_w \times A_p$$

where:

A = surface-water area in km^2

N_w = number of water pixels

A_p = area of one pixel in km^2

Therefore:

$$A = N_w \times 0.0001$$

Annual absolute change was calculated as:

$$\Delta A = A_t - A_{t-1}$$

where:

A_t = current-year water area

A_{t-1} = previous-year water area

Annual percentage change was calculated as:

$$\% \Delta A = [(A_t - A_{t-1}) / A_{t-1}] \times 100$$

Time-series Analysis

A line graph was produced from Ms. Excel to show the temporal variation in surface-water extent from 2022 to 2026.

A linear trendline was fitted to the five annual observations. The coefficient of determination was calculated using:

$$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$$

The resulting R^2 value was used to describe the proportion of variation represented by the fitted linear trend. The trend analysis was treated as descriptive because only five annual observations were available.

Spatial Change Detection

Spatial change detection was performed using the final Majority Filter water masks.

For the overall 2022–2026 comparison, the two binary masks were combined using:

$$PCC = 2 \times M_{2022} + M_{2026}$$

Where:

M_{2022} = final 2022 water mask

M_{2026} = final 2026 water mask

The resulting classes were interpreted as:

0 = stable non-water

1 = water gain

2 = water loss

3 = persistent water

The area of each transition class was calculated from the number of pixels and the 10 m pixel area.

The same transition procedure was designed for the consecutive periods:

2022–2023

2023–2024

2024–2025

2025–2026

Results

Annual MNDWI Water Delineation

The validated MNDWI approach produced spatially coherent water masks for all five years. The final masks showed the reservoir as the dominant water feature within the study area, with changes concentrated around the reservoir margins and connected arms. The raw and Majority Filter water counts are shown in Table 4.

Table 4. Annual water-mask pixel counts before and after Majority Filtering

Year	Raw non-water pixels	Raw water pixels	Filtered non-water pixels	Filtered water pixels
2022	18,268	33,104	18,273	33,097
2023	18,477	32,895	18,424	32,929
2024	6,661	44,711	6,312	44,854
2025	13,776	37,596	13,755	37,591
2026	9,351	42,021	9,160	42,103

Source: Author, 2026.

The Majority Filter produced relatively small changes in the mapped water pixels. The 2022 mask decreased by 7 water pixels, the 2023 mask increased by 34, the 2024 mask increased by 143, the 2025 mask decreased by 5, and the 2026 mask increased by 82.

Annual Surface-Water Area

The final surface-water areas derived from the Majority Filter masks are presented in Table 4.

Table 5. Annual surface-water extent of Kopa Dam, 2022–2026

Year	MNDWI threshold	Final-water pixels	Water-area (km ²)	Annual-change (km ²)	Annual change (%)
2022	−0.01	33,097	3.3097	—	—
2023	−0.01	32,929	3.2929	−0.0168	−0.51
2024	−0.18	44,854	4.4854	+1.1925	+36.21
2025	−0.01	37,591	3.7591	−0.7263	−16.19
2026	−0.25	42,103	4.2103	+0.4512	+12.00

Source: Author, 2026.

The minimum mapped water area was 3.2929 km² in 2023. The maximum was 4.4854 km² in 2024. The difference between the minimum and maximum annual water areas was 1.1925 km².

The mean water area across the five observations was 3.8115 km², with a sample standard deviation of approximately 0.533 km².

Temporal Trend of Surface-Water Area

The annual water-area series showed a fluctuating pattern. Water area decreased slightly from 3.3097 km² in 2022 to 3.2929 km² in 2023. It then increased sharply to 4.4854 km² in 2024. The area decreased to 3.7591 km² in 2025 and increased again to 4.2103 km² in 2026. The linear trendline showed an overall positive tendency. The fitted trend was:

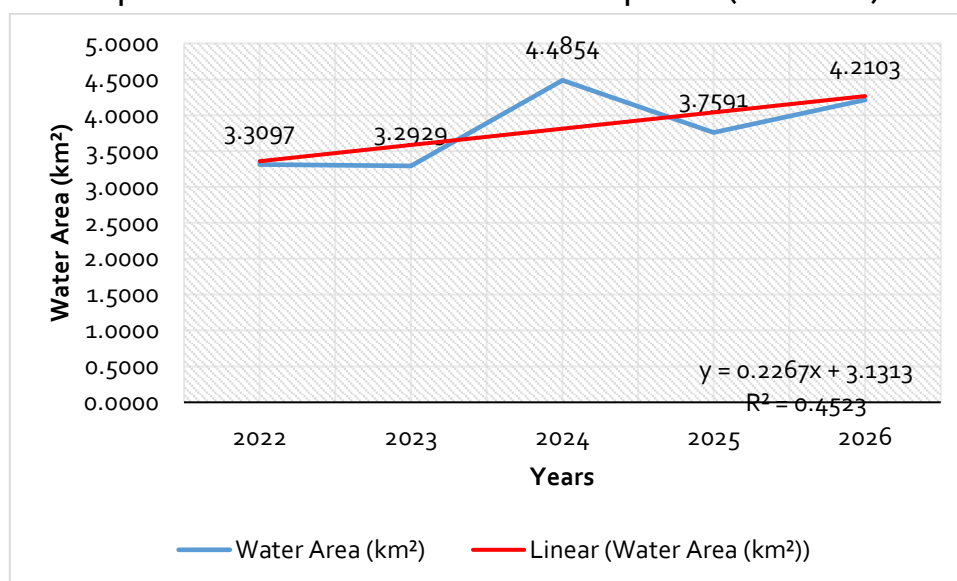
$$y = 0.2267x + 3.1313$$

The coefficient of determination was:

$$R^2 = 0.4523$$

The trend therefore represented an increasing tendency across the five observations, although the observed values showed substantial year-to-year fluctuation.

Figure 2: Temporal Trend of Surface-Water Area of Kopa Dam (2022–2026)



Source: Author, 2026.

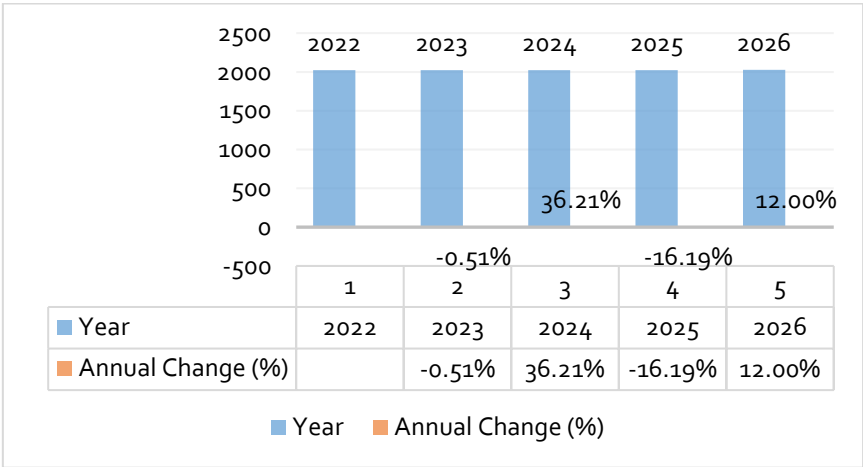
Annual Surface-Water Changes

The smallest annual change occurred between 2022 and 2023, when water area declined by 0.0168 km², equivalent to 0.51%. The largest increase occurred between 2023 and 2024, where Water area increased by 1.1925 km², equivalent to 36.21%. The largest decline occurred between 2024 and 2025. Water area decreased by 0.7263 km², equivalent to 16.19%.

Between 2025 and 2026, the water area increased by 0.4512 km², equivalent to 12.00%.

Between 2022 and 2026, the water area increased from 3.3097 km² to 4.2103 km².
The absolute increase was:
 $\Delta A = 4.2103 - 3.3097$
 $\Delta A = 0.9006 \text{ km}^2$
The overall percentage increase was:
 $\% \Delta A = [(4.2103 - 3.3097) / 3.3097] \times 100$
 $\% \Delta A = 27.21\%$
Thus, the final annual-area analysis recorded an overall increase of 0.9006 km², or 27.21%, across the study period.

Figure 3: Annual Percentage Change in Surface-Water Area of Kopa Dam (2022–2026).



Source: Author, 2026.

Spatial Water Gain and Loss

The overall transition analysis produced four classes: stable non-water, water gain, water loss, and persistent water.

Table 6: Overall surface-water transitions between 2022 and 2026

Transition class	Pixel count	Area (ha)	Area (km²)
Stable non-water	8,563	85.63	0.8563
Water gain	9,546	95.46	0.9546
Water loss	564	5.64	0.0564
Persistent water	32,531	325.31	3.2531
Total	51,204	512.04	5.1204

Source: Author, 2026.

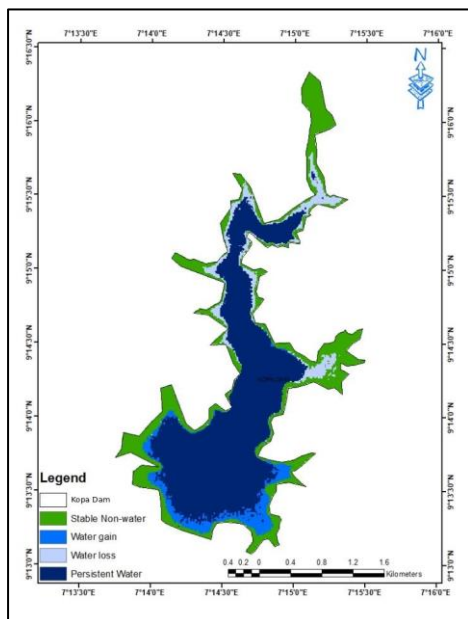
The spatial transition analysis showed that water gain was considerably greater than water loss. Water gain covered 95.46 ha, whereas water loss covered 5.64 ha. Persistent water occupied 325.31 ha.
The net transition derived from the gain and loss classes was:

Net change = $0.9546 - 0.0564$

Net change = 0.8982 km^2

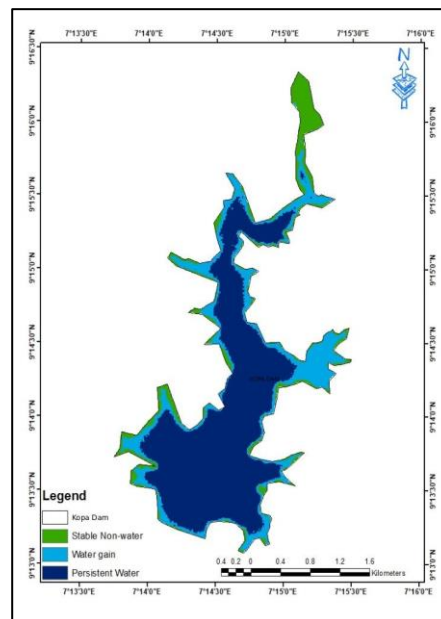
The transition map therefore showed a strong dominance of water-gain areas over water-loss areas during the 2022–2026 comparison.

Figure 4: 2022–2023 change map.



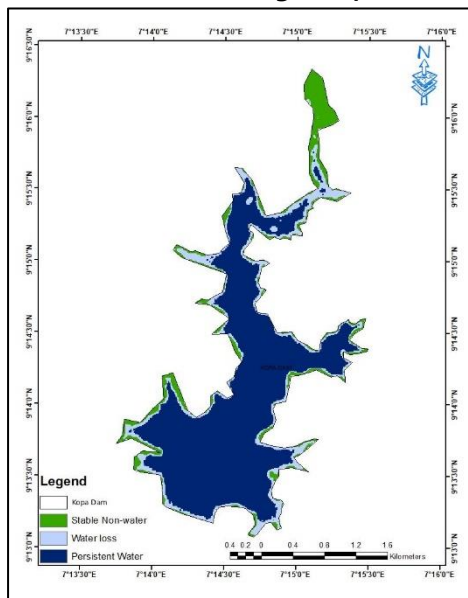
Source: Author, 2026.

Figure 5: 2023–2024 change map.



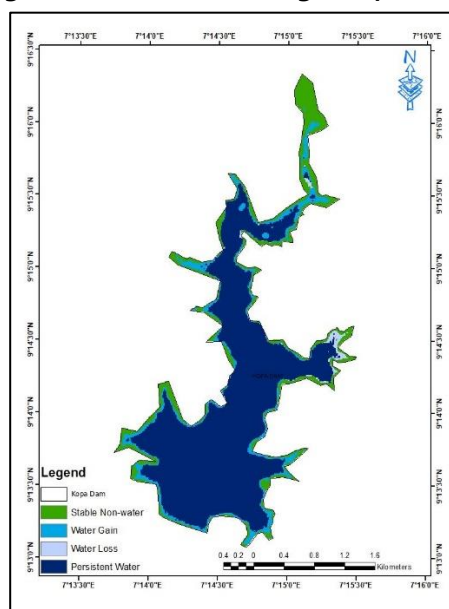
Source: Author, 2026.

Figure 6: 2024–2025 change map.



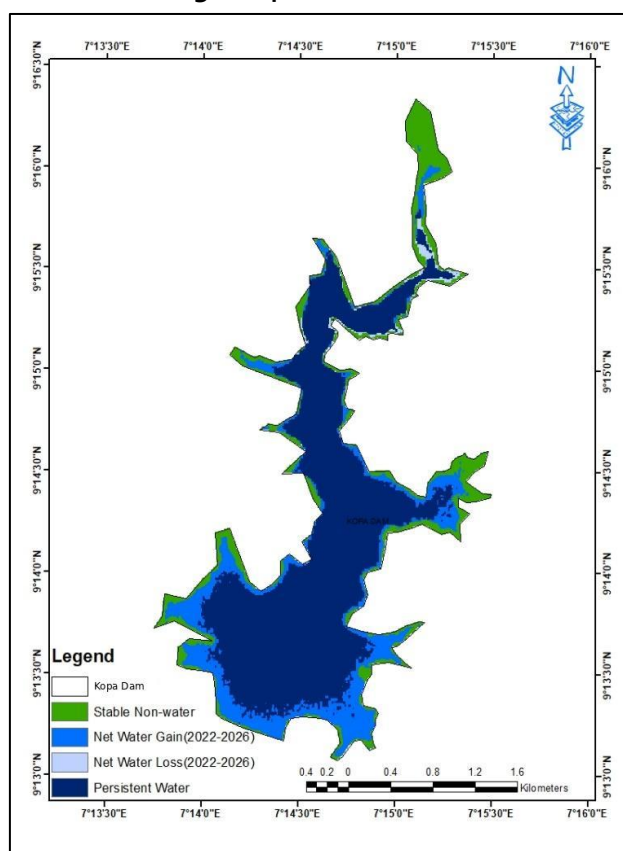
Source: Author, 2026.

Figure 7: 2025–2026 change map.



Source: Author, 2026.

Figure 8: Overall 2022–2026 change map.



Source: Author, 2026.

The direct annual-area comparison produced a 0.9006 km² increase, whereas the transition raster produced a net transition gain of 0.8982 km². The difference is 0.0024 km², equivalent to 24 pixels.

Discussion

The results show that Kopa Dam experienced clear year-to-year variation in surface-water extent between January 2022 and January 2026. The analysis was based entirely on Sentinel-2 Level-2A imagery and MNDWI-derived water masks. The workflow produced five annual water masks, which were subsequently refined using Majority Filtering before area calculation and temporal analysis. The results therefore provide a consistent spatial record of the mapped surface-water extent across the five observation years.

MNDWI-Based Surface-Water Delineation

MNDWI provided the basis for all surface-water delineation in this study. The index was calculated from the Sentinel-2 green band (B03) and shortwave infrared band (B11). The resulting MNDWI rasters showed different value distributions among the five years. This variation was important because the same numerical threshold did not produce the same level of water and non-water separation across all images.

The workflow therefore used reference points to examine the MNDWI values of known water and non-water locations. This process provided the basis for selecting an appropriate threshold for each image. The final thresholds were -0.01 for 2022, -0.01 for 2023, -0.18 for 2024, -0.01 for 2025, and -0.25 for 2026.

The results demonstrate why the threshold was treated as an image-specific classification parameter rather than a fixed value for the entire study period. The 2022, 2023, and 2025 images were successfully classified using -0.01 . The 2024 image required a lower threshold of -0.18 . The 2026 image required an even lower threshold of -0.25 .

The 2024 validation was particularly important in the workflow. The initial -0.01 threshold did not provide the required separation for the reference data. Additional threshold testing was therefore performed. The -0.18 threshold produced the required classification pattern and was retained for the final 2024 water mask.

The 2026 validation also showed a clear separation between the two reference classes. The 80 Class 1 points had MNDWI values from -0.225540 to -0.073065 , with a mean of -0.139579 . The 71 Class 2 points had values from -0.431180 to -0.285296 , with a mean of -0.351135 . The selected threshold of -0.25 therefore separated the two observed value ranges and produced a total of 151 correctly classified reference points in the threshold assessment.

The threshold-selection process strengthened the final water masks because classification decisions were based on the observed spectral behavior of each image rather than on an assumed common threshold. This image-specific approach is also consistent with evidence that water-index performance can vary across environmental settings.

Effect of Majority Filtering on the Water Masks

The raw binary water masks contained small isolated pixels and minor spatial irregularities. The Majority Filter was therefore applied before the final area calculation. The purpose was to improve the spatial coherence of the classified water surface and reduce isolated classification noise.

The effect of the filter was relatively small for most years. In 2022, the number of water pixels changed from 33,104 to 33,097. In 2023, it changed from 32,895 to 32,929. The 2024 water-pixel count changed from 44,711 to 44,854. In 2025, it changed from 37,596 to 37,591, while 2026 changed from 42,021 to 42,103.

These changes indicate that the Majority Filter did not substantially alter the overall mapped water extent. Instead, it made small local corrections to the binary masks. This was important because the final area statistics were based on the filtered masks rather than the raw classifications.

The filtered masks also provided a cleaner basis for spatial comparison between years. The same filtering approach was applied across the five annual masks so that the final products remained comparable.

Annual Surface-Water Extent

The final surface-water areas showed considerable inter-annual variation. Kopa Dam had a mapped water area of 3.3097 km² in 2022. This decreased slightly to 3.2929 km² in 2023. The water area then increased sharply to 4.4854 km² in 2024. It declined to 3.7591 km² in 2025 before increasing again to 4.2103 km² in 2026.

The 2023 value was the smallest mapped water area during the study period. The 2024 value was the largest. The difference between these two years was 1.1925 km².

The annual sequence can therefore be expressed as:

$$2022 = 3.3097 \text{ km}^2$$

$$2023 = 3.2929 \text{ km}^2$$

$$2024 = 4.4854 \text{ km}^2$$

$$2025 = 3.7591 \text{ km}^2$$

$$2026 = 4.2103 \text{ km}^2$$

The pattern does not represent continuous annual expansion. Instead, the reservoir showed a small reduction in 2023, a major expansion in 2024, a substantial contraction in 2025, and a subsequent increase in 2026.

Annual Percentage Change in Surface-Water Area

The annual percentage analysis provided a clearer measure of the magnitude of change between consecutive observations. Between 2022 and 2023, the water area decreased by 0.0168 km², representing a -0.51% change; between 2023 and 2024, the water area increased by 1.1925 km², representing a +36.21% change. This was the largest annual increase recorded in the study; between 2024 and 2025, the water area decreased by 0.7263 km², representing a -16.19% change. This was the largest annual decline; between 2025 and 2026, the water area increased by 0.4512 km², representing a +12.00% change.

The annual percentage results therefore confirm the strong fluctuation observed in the mapped water areas. The 2024 expansion was followed by a substantial reduction in 2025, after which the water surface recovered in 2026.

The direct comparison between the first and final observation shows an increase in mapped surface-water area.

The overall change was calculated as:

$$\Delta A = A_{2026} - A_{2022}$$

$$\Delta A = 4.2103 - 3.3097$$

$$\Delta A = 0.9006 \text{ km}^2$$

The percentage change was calculated as:

$$\% \Delta A = [(A_{2026} - A_{2022}) / A_{2022}] \times 100$$

$$\% \Delta A = [(4.2103 - 3.3097) / 3.3097] \times 100$$

$$\% \Delta A = 27.21\%$$

The mapped surface-water area therefore increased by 0.9006 km², equivalent to 27.21%, between January 2022 and January 2026.

This overall increase should be interpreted together with the annual fluctuations. The final 2026 water area was larger than the 2022 water area, but the reservoir did not increase continuously throughout the study period. The largest temporary expansion occurred in 2024, followed by a decline in 2025.

Temporal Trend of Surface-Water Area

The time-series analysis provided an overall measure of the direction of change across the five annual observations. The linear trendline produced the equation:

$$y = 0.2267x + 3.1313$$

The coefficient of determination was:

$$R^2 = 0.4523$$

The positive slope indicates an overall increasing tendency in mapped surface-water area across the study period. The R^2 value of 0.4523 indicates that the fitted linear relationship explains 45.23% of the variation in the five annual observations.

The trend should be interpreted as an overall tendency rather than evidence of continuous annual growth. The observed values moved in different directions across the study period. Water area decreased slightly from 2022 to 2023, increased strongly in 2024, declined in 2025, and increased again in 2026.

Spatial Distribution of Water Gain and Loss

The temporal statistics show how much the water area changed. The spatial transition analysis shows where the change occurred.

The 2022 and 2026 final water masks were compared to produce four transition classes:

0 = stable non-water

1 = water gain

2 = water loss

3 = persistent water

The overall transition analysis produced the following pixel counts:

Stable non-water = 8,563 pixels

Water gain = 9,546 pixels

Water loss = 564 pixels

Persistent water = 32,531 pixels

Using the 10 m × 10 m pixel size, the corresponding areas were:

Stable non-water = 85.63 ha

Water gain = 95.46 ha

Water loss = 5.64 ha

Persistent water = 325.31 ha

The gain area was therefore substantially larger than the loss area. The difference was:

$$\begin{aligned} \text{Water gain} - \text{Water loss} &= 95.46 - 5.64 \\ &= 89.82 \text{ ha} \end{aligned}$$

This indicates that the overall spatial transition from 2022 to 2026 was dominated by expansion of the mapped water surface.

Persistent water also represented the largest transition category. This shows that a substantial portion of the reservoir remained classified as water in both 2022 and 2026.

Relationship Between the Area Statistics and Spatial Transition

The annual area statistics and the spatial transition analysis provide complementary information. Similar geospatial monitoring approaches have been used to support water-accounting and management where continuous field observations are limited (Ngebe et al., 2022).

The annual statistics show that the mapped water area increased from 3.3097 km² in 2022 to 4.2103 km² in 2026. The transition analysis shows that this overall increase was spatially dominated by water-gain pixels.

The direct annual-area calculation produced an overall increase of 0.9006 km². The transition analysis produced 0.9546 km² of water gain and 0.0564 km² of water loss. The net transition change was therefore:

$$0.9546 - 0.0564 = 0.8982 \text{ km}^2$$

The difference between the direct annual-area changes and the transition-based net change is:

$$0.9006 - 0.8982 = 0.0024 \text{ km}^2$$

This difference represents 24 pixels at the 10 m resolution. It is small relative to the total mapped area and reflects the difference between the direct annual-area calculation and the raster transition accounting.

Implications of the Observed Water-Area Pattern

The five-year record shows that Kopa Dam did not maintain a constant surface-water extent. The mapped area changed substantially between successive January observations. The wider Suleja area has also been identified as vulnerable to flood risk, with recent work linking flood exposure to rainfall, land use, terrain and hydrological factors (Umar et al., 2026).

The most important change occurred between 2023 and 2024, when the mapped water surface increased by 1.1925 km². The following year recorded a reduction of 0.7263 km². The increase in 2026 indicates a partial recovery from the 2025 reduction. A plausible explanation for part of the 2024 expansion is increased antecedent wet-season inflow following the rainy season. This interpretation is supported only indirectly by reports of widespread flooding and heavy rainfall across Niger State during 2024 (International Organization for Migration [IOM], 2024), rather than by measurements at Kopa Dam itself. Conversely, the contraction from 2024 to 2025 could reflect lower antecedent inflow, reservoir drawdown or abstraction, evaporation, or other operational and catchment processes. These factors were not measured in the present study and therefore cannot be

identified as causal drivers. The observed changes should consequently be treated as evidence of surface-water fluctuation, with hydrological and operational explanations regarded as plausible interpretations requiring field-based verification.

The study therefore identifies three important characteristics of the reservoir during the observation period. First, the water surface remained spatially persistent across a large part of the reservoir. Second, the reservoir experienced substantial temporary expansion and contraction. Third, the final mapped water area in 2026 exceeded the 2022 baseline.

The analysis measures surface-water extent. It does not directly measure reservoir depth, storage volume, inflow, rainfall, evaporation, abstraction, or sediment volume. The results should therefore be interpreted as changes in mapped surface-water area rather than direct changes in reservoir storage.

The absence of field bathymetric or water-level observations in this study also means that the study does not convert the mapped surface area into reservoir volume. Such an analysis would require independent elevation or depth information and an established relationship between water level, area, and storage.

Comparison with Previous Studies

The present results are consistent with previous evidence that reservoir surface-water extent can fluctuate substantially between observation years. Frichi et al. (2023), in their study of Kiri Dam in Adamawa State, Nigeria, reported substantial inter-annual changes in surface-water area and identified MNDWI as the most suitable of the tested indices. Their observation of a non-monotonic pattern, including an increase within an otherwise declining sequence, is comparable to the fluctuations observed at Kopa Dam, although the direction and magnitude of change differed between the two reservoirs.

For Kopa Dam, Samaila-Ija et al. (2014) reported a surveyed surface area smaller than the designed area and a substantial difference between the designed and observed water volumes, which they associated with sedimentation and siltation. This earlier bathymetric evidence complements the present surface-area analysis by demonstrating that mapped surface-water extent alone cannot be used to infer changes in reservoir storage volume. Such interpretation requires corresponding water-level observations and information on reservoir bathymetry and geometry. Ngebe et al. (2022) similarly demonstrated that Sentinel-2 imagery can support cost-effective dam water-level estimation when combined with elevation and in-situ observations.

At a broader scale, Ogilvie et al. (2025) showed that remote sensing can reveal pronounced inter-annual surface-water variability where conventional hydrological observations are limited. Recent evaluations of Sentinel-2 water indices have also shown that index performance can vary with environmental conditions (Tesfaye & Breuer, 2024; Yeboah et al., 2026). Overall, these findings support the use of a consistent image-based monitoring framework for Kopa Dam while emphasizing that surface-water changes should be interpreted alongside hydrological, operational, and bathymetric information.

Conclusion

This study assessed the spatio-temporal changes in the surface-water extent of Kopa Dam from January 2022 to January 2026 using Sentinel-2 Level-2A imagery, MNDWI, and ArcGIS 10.8. MNDWI was used as the only spectral water index. Image-specific threshold validation produced thresholds of -0.01 for 2022, -0.01 for 2023, -0.18 for 2024, -0.01 for 2025, and -0.25 for 2026. The threshold variation demonstrates that the five images required separate classification decisions based on their observed MNDWI distributions.

The final Majority-Filtered water masks produced water areas of 3.3097 km^2 in 2022, 3.2929 km^2 in 2023, 4.4854 km^2 in 2024, 3.7591 km^2 in 2025, and 4.2103 km^2 in 2026. The smallest mapped area occurred in 2023, while the largest occurred in 2024. The largest annual increase occurred between 2023 and 2024, with an increase of 1.1925 km^2 or 36.21% . The largest annual decline occurred between 2024 and 2025, with a reduction of 0.7263 km^2 or 16.19% . Between 2025 and 2026, the water surface increased by 0.4512 km^2 or 12.00% .

The direct comparison of the first and final observations showed an increase of 0.9006 km^2 , equivalent to 27.21% . The linear trendline showed a positive overall tendency with $R^2 = 0.4523$, although the annual observations demonstrated substantial fluctuations rather than continuous growth. The spatial transition analysis provided additional evidence of the overall change. Between 2022 and 2026, 95.46 ha were classified as water gain, 5.64 ha as water loss, 325.31 ha as persistent water, and 85.63 ha as stable non-water. Water gain therefore greatly exceeded water loss during the overall comparison.

This study provides a practical method for monitoring Kopa Dam using freely available Sentinel-2 imagery and GIS. It combines image-specific MNDWI threshold validation, binary classification, spatial filtering, area calculation, temporal analysis, and spatial transition mapping. The resulting dataset establishes a five-year January baseline of surface-water extent for Kopa Dam and provides a basis for continued monitoring using the same processing framework. The Niger State Water Board and other agencies responsible for Kopa Dam management should integrate this satellite-based monitoring with routine field water-level measurements and periodic bathymetric surveys. These combined observations can support development and updating of an area–volume–elevation relationship for the reservoir and improve assessment of water-supply reliability and storage conditions.

Recommendations

The Niger State Water Board and the authorities responsible for Kopa Dam should establish routine field water-level observations that can be paired with the Sentinel-2 MNDWI workflow used in this study. Periodic bathymetric surveys should also be undertaken to update the reservoir bed model and establish an area–volume–elevation relationship. This would allow future surface-area observations to be related more directly to water level and storage volume. For operational monitoring, the same January image-selection framework can be maintained for annual baseline comparison, while additional images during the wet and dry seasons can be used to capture short-term fluctuations. Future studies should also

incorporate rainfall, inflow, abstraction, evaporation, reservoir-operation records, and land-use information to test the drivers of observed annual changes rather than inferring causality from surface-water area alone.

Conflict of Interest

The author(s) declare(s) that there is no conflict of interest regarding the publication of this article.

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Author's Contribution (CRediT)

Musa, A.: Conceptualisation, Methodology, Writing – Original Draft, Data Curation & Analysis. Dauda, W.A.: Supervision, Writing – Review & Editing. Sunusi, A.: Validation. Abubakar, Z.I: Investigation.

Data Availability

The datasets generated and analysed during the current study are available from the corresponding author on reasonable request.

Disclosure Statement

The authors used ChatGPT (OpenAI) for language editing and grammar correction of the manuscript. The authors remain responsible for the study design, data, analysis, interpretation, and final manuscript content.

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